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SIGNOVA- SIGN LANGUAGE TRANSLATOR

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Abstract

Signova is a web-based sign language recognition system designed to recognize a selected set of hand gestures and convert them into readable words.

The system uses webcam input, MediaPipe hand landmarks, Python-based recognition, Flask, React and MySQL.

The current prototype focuses on a limited vocabulary and provides a foundation that can later be extended to a larger sign language translation system.



Problem Statement

Communication between sign-language users and people unfamiliar with sign language can be difficult.

Existing communication often requires:

- A person who understands sign language
- Manual interpretation
- Additional communication tools

Problem:

Develop a simple computer-vision-based system that can recognize selected sign-language gestures from webcam input and display the corresponding words.



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Motivation



- Reduce communication barriers.
- Explore the application of computer vision in sign-language recognition.
- Provide an accessible software-based prototype.
- Create a system that can work with a normal webcam.
- Build a foundation for future real-time sign-language translation.



Objectives



- Capture hand gestures using a webcam.
- Detect hand landmarks using MediaPipe.
- Convert landmarks into numerical features.
- Recognize selected sign-language gestures.
- Display the recognized word through a web interface.
- Store sign information and recognition history using MySQL.
- Develop a system that can be expanded with additional signs.

Research Gap

Identified gaps:

- Limited vocabulary in many prototype systems.
- Difficulty recognizing visually similar gestures.
- Sensitivity to lighting, hand position and camera quality.
- Need for simple and affordable implementations.
- Need for systems that combine recognition with a usable web interface and database.

Our focus:

A lightweight prototype combining hand landmark detection + gesture recognition + web interface + database storage.



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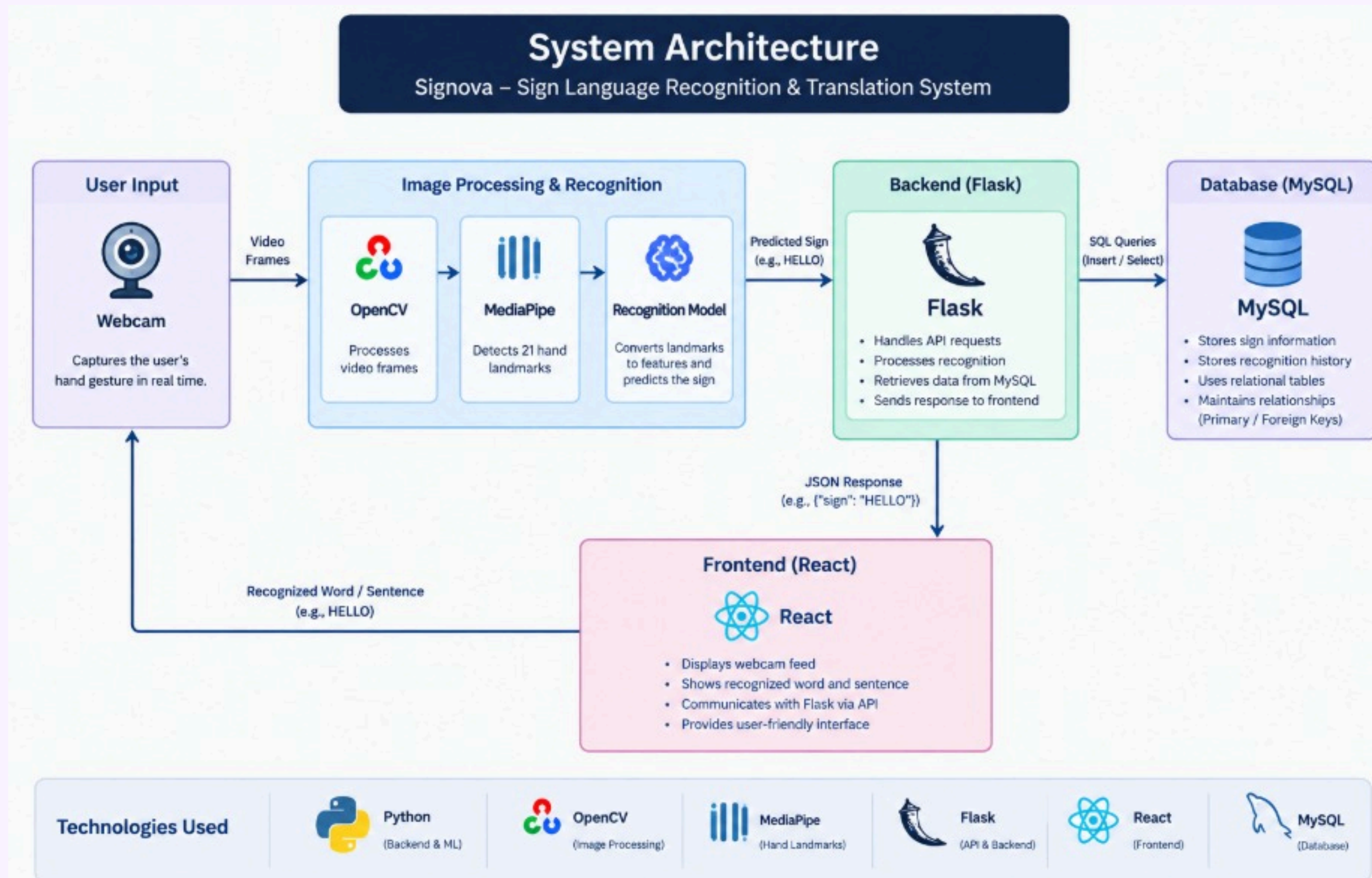
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Proposed Methodology



System Architecture



Technologies Used

Frontend

React

HTML

CSS

JavaScript

Backend

Python

Flask

Computer Vision

MediaPipe

OpenCV

Machine Learning / Recognition

Landmark-based feature extraction

Nearest-neighbor based recognition

Database

MySQL

Development

VS Code

Python virtual environment

GitHub



CO's *Implementation*

CO1-

Hand Detection & Sign Recognition

Webcam & Image Processing

- Webcam captures the user's hand gestures.
- OpenCV processes the live video frames.
- Frames are passed to the hand detection module.

Hand Landmark Detection

- MediaPipe detects 21 hand landmarks.
- Each landmark provides X, Y and Z coordinates.
- The landmarks are normalized to reduce the effect of hand position and size.

Sign Recognition

- Extracted landmarks are converted into numerical features.
- Features are provided to the trained recognition model.
- The model predicts the corresponding sign.



CO's *Implementation*

CO2-

Flask Backend & MySQL Database

Flask Backend:

- Flask acts as the communication layer between the frontend and recognition system.
- Provides REST API endpoints for recognition and sign information.
- The /recognize API receives the input and returns the predicted sign.

MySQL Database:

- Stores sign-related information.
- Stores recognition history.
- Uses relational tables with primary and foreign keys.
- SQL queries are used to retrieve and manage data.



CO's Implementation

CO3-

React Frontend & Sentence Recognition

React Frontend

- Provides the main user interface.
- Displays the webcam and recognition results.
- Communicates with the Flask backend through APIs.

Word Recognition

- Receives the predicted sign from the backend.
- Converts the predicted label into a readable word.
- Displays the result to the user.

Sentence Recognition

- Collects multiple recognized words.
- Combines the words into a readable sentence.
- Provides the foundation for future improvements in sentence formation.



SCREENSHOTS OF WORKING DEMO

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BRIDGING COMMUNICATION
THROUGH TECHNOLOGY

Camera Camera Input

Offline

Camera is off

Start the camera to begin recognition.

Start Camera

Start Camera

Recognize Now

Latest recognition: Please

Supported Signs

View All

Detected Sign

Please

Recognized sign from the current model.

Meaning

Recognized sign

Current Sentence

Clear

Hello Please

Translation History

View All

02 Oct 2026, 07:27 pm

Please

02 Oct 2026, 07:27 pm

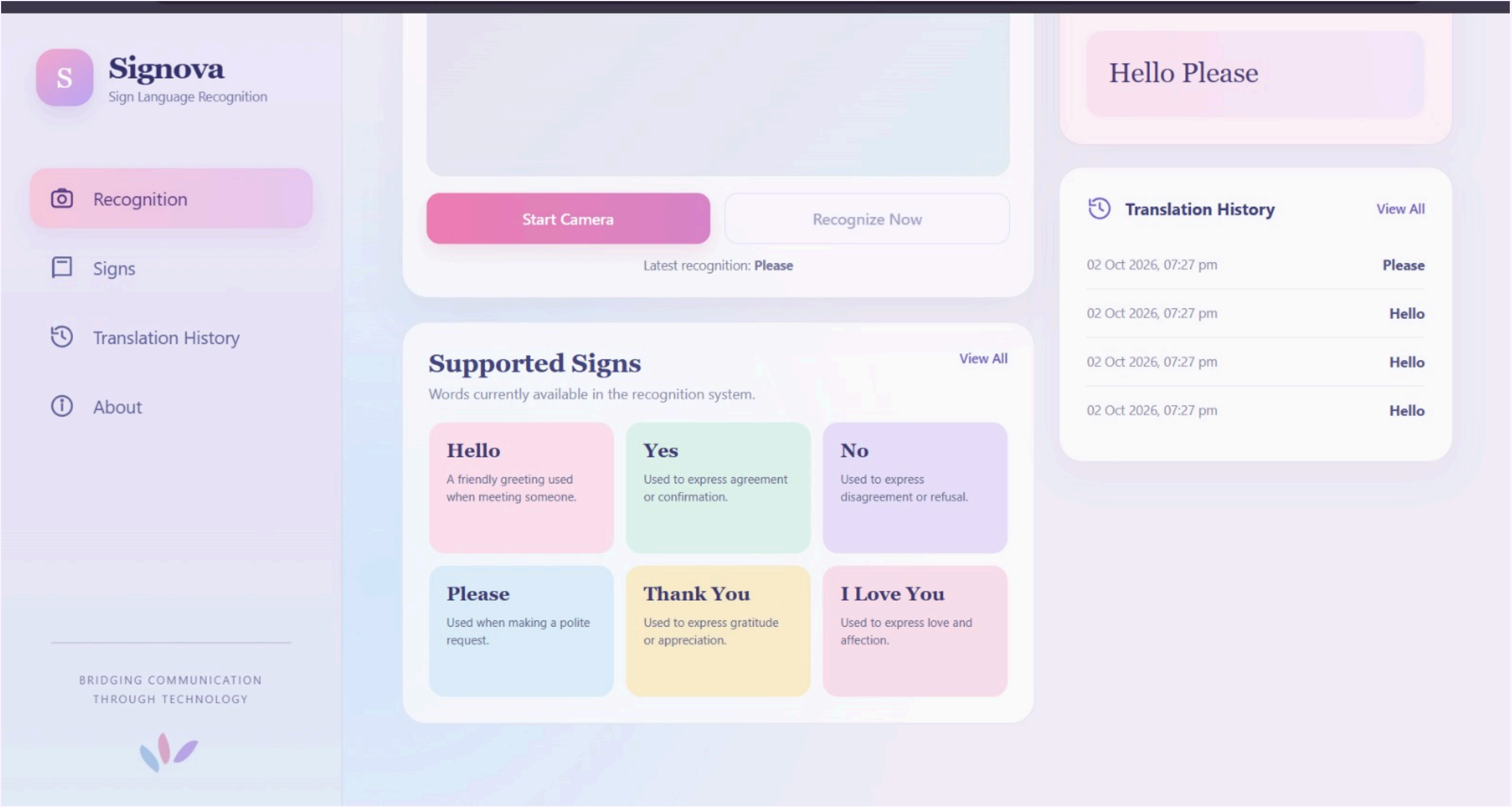
Hello

02 Oct 2026, 07:27 pm

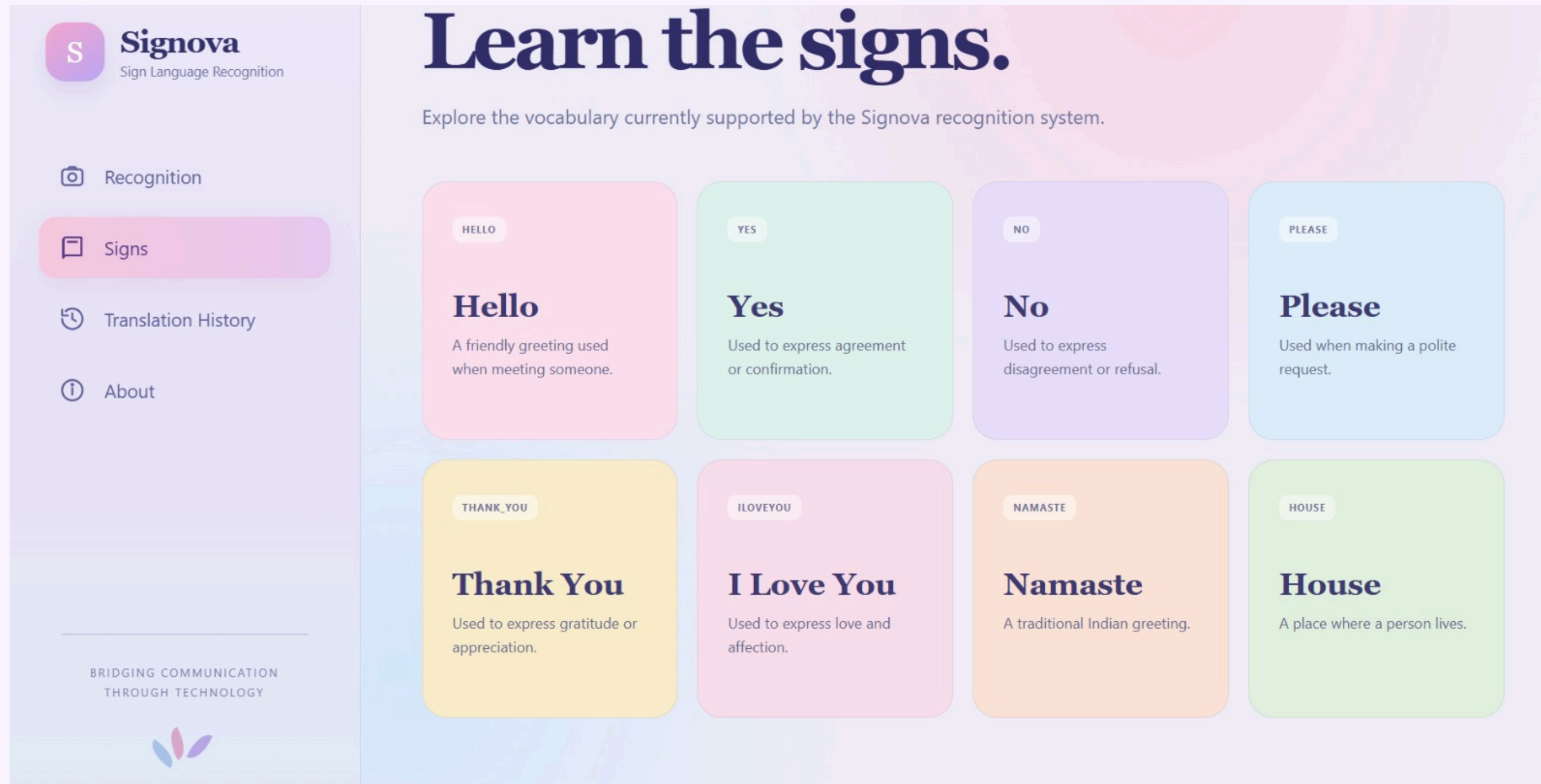
Hello



SCREENSHOTS OF WORKING DEMO



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TRANSLATION RECORD

Translation History

Previously recognized signs are listed here for reference.

Recent Translations

9 translations recorded

Clear History

TIME	DETECTED SIGN	MEANING
02 Oct 2026, 07:27 pm	Please	Recognized sign
02 Oct 2026, 07:27 pm	Hello	Recognized sign
02 Oct 2026, 07:27 pm	Hello	Recognized sign
02 Oct 2026, 07:27 pm	Hello	Recognized sign
02 Oct 2026, 07:27 pm	Hello	Recognized sign
02 Oct 2026, 07:27 pm	Hello	Recognized sign

BRIDGING COMMUNICATION
THROUGH TECHNOLOGY

Comparison with Existing System



Feature	Existing System	Proposed System
Input	May require dedicated setup	Standard webcam
Recognition	Standalone / limited integration	Landmark-based recognition
Database	May be separate	MySQL integrated
Interface	Varies by system	React web interface
History	Often limited	Recognition history stored



Future Work

- Expand the sign vocabulary.
- Improve recognition accuracy.
- Add dynamic/continuous gesture recognition.
- Improve sentence formation.
- Add voice output.
- Improve robustness to lighting and background changes.
- Explore more advanced machine-learning/deep-learning models.
- Deploy the system as a more complete real-time application.



Conclusion

Signova demonstrates a basic real-time sign-language recognition system using computer vision and web technologies. The system extracts hand landmarks using MediaPipe, recognizes selected gestures using a landmark-based recognition model, and integrates Flask, React and MySQL. The current prototype provides a foundation that can be extended with a larger vocabulary, improved recognition models and continuous sign-language translation.





References

- [MediaPipe documentation — Hand Landmarker](#)
- [OpenCV documentation](#)
- [Flask documentation](#)
- [React documentation](#)
- [MySQL documentation](#)
- [Hugging Face — vedit031/isolated-40words dataset](#)
- [Relevant research papers from your literature survey](#)

